

Comparative Analysis of Deep Learning Models for Intracerebral Hemorrhage Detection from Computed Tomography

Taiseer Rakiin Ahad¹, Saadia Binte Alam², Farhan Ha-mim³, Ashad Nur⁴, Sefatul Wasi⁵ and Rashedur Rahman⁶

^{1,2,3,4,5,6}Department of Computer Science and Engineering

Independent University Bangladesh, Dhaka, Bangladesh

Email- taiseerrakiin@gmail.com, saadiabintealam@gmail.com, farhanhamim2001@gmail.com, ashadnur1996@gmail.com, Sefatulwasi@gmail.com, rashed.riyadh14@gmail.com

Abstract— Intracerebral hemorrhage (ICH) is a serious medical condition caused by bleeding within the brain tissue, which can lead to life-threatening complications if not diagnosed quickly. This study compares the performance of three advanced deep learning models: InceptionV3, DenseNet201 and EfficientNetB4 for detecting ICH from computed tomography (CT) scans. The novelty of this work lies in the use of preprocessing techniques such as windowing with Hounsfield Units and Skull segmentation to provide cleaner input data, along with Grad-CAM visualizations to ensure the model's predictions are interpretable. Using comprehensive evaluations, including metrics like AUROC, precision and F1 score, EfficientNetB4 achieved the best results with an accuracy of 0.85, precision of 0.87 and F1-score of 0.86, demonstrating its potential to assist medical professionals in making faster and more reliable diagnoses.

Keywords—Intracerebral hemorrhage, Deep learning, Computed tomography, Image classification, InceptionV3, DenseNet201, EfficientNetB4, Medical imaging, Computer-aided diagnosis

I. INTRODUCTION

Intracerebral hemorrhage (ICH) is caused by abnormal bleeding within the brain tissue [1], which increases pressure in the brain, obstructs blood flow and reduces oxygen supply. This can cause damage to brain cells, leading to severe outcomes such as loss of consciousness, disability, or even death [2]. Advanced imaging technologies like Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) are crucial for diagnosing ICH, with procedures such as craniotomy, hematoma evacuation, or aneurysm cutting often used as treatment [3]. Additional complications can include hydrocephalus and persistent neurological issues, which can impair brain function and increase infection risks like meningitis or brain abscesses [4]. The global incidence of ICH is rising, mainly due to aging and increasing cases of hypertension [5]. ICH accounts for 10 to 15% of all stroke cases [6], with varying regional estimates of 8% to 15% in countries like the United States, United Kingdom, and Australia, and 18% to 24% in East Asian countries like Japan and Korea [5].

The occurrence of ICH varies by country and ethnicity, with a mortality rate of 40% to 50% within 30 days [7].

Survivors often face a 50-70% morbidity rate, experiencing long-term issues like motor impairment and chronic discomfort [5]. Diagnosing ICH can be a challenging task due to symptom variability, limited imaging capabilities [7] and factors such as age, pre-existing health conditions, and limited access to specialists. Early diagnosis is crucial to avoid severe consequences like increased bleeding and infections, including complications like hydrocephalus [8]. Automated or assistive methods, such as machine learning algorithms, show promise in improving early detection of hemorrhages.

Previous research has demonstrated the effectiveness of deep learning algorithms in detecting brain bleeding from CT scans, while also highlighting several limitations. Key challenges include the need for large datasets, variations in image quality, and reliance on expert annotations. For example, Grewal et al. [9] developed RADnet, a deep learning model achieving 93.5% accuracy in detecting hemorrhages, but it required a substantial dataset of 12,000 CT scans. Similarly, Lee et al. [10] designed a deep learning method that, despite being constrained by image quality variability, achieved a sensitivity of 95.5%. Chilamkurthy et al. [11] introduced a deep learning model with 92.5% accuracy in identifying major findings in head CT scans, but it required professional annotation. Another study employed a 3D joint convolutional and recurrent neural network which, while hindered by the need for large datasets, detected ICH with 94.5% accuracy [12]. Furthermore, a deep learning model in [13] demonstrated the ability to identify ICH with 96.5% accuracy, despite its dependency on expert annotation.

In this paper, we compare three advanced deep learning models: InceptionV3, DenseNet201, and EfficientNetB4 for detecting ICH based on three classifications. To improve the quality of the data, we used techniques like windowing with Hounsfield Units and skull segmentation, and we also removed distorted images to avoid confusing the models. These steps, combined in our study, set our approach apart from previous studies. We also used Grad-CAM visualizations to make the models' predictions more reliable, ensuring they focused on the right areas in the scans. These models were trained on the RSNA Intracranial Hemorrhage

dataset, which is publicly available on Kaggle [14]. Our analysis also considered practical factors like computational efficiency and the pros and cons of the different models, providing valuable insights for future researchers and making these tools more applicable in real-world medical settings.

II. DATASET

The RSNA Intracranial Hemorrhage dataset includes a variety of CT slices categorized into five types of hemorrhages: Epidural Hematoma, Intraparenchymal Hemorrhage, Intravascular Hemorrhage, Subdural Hematoma, and Subarachnoid Hemorrhage. It also contains a CSV file with labels and a unique ID for each CT slice, including an "any" label which further specifies the existence of any form of hemorrhage in the CT slice.

The dataset consists of 752,602 Digital Imaging and Communications in Medicine (DICOM) files, where each file corresponds to a single slice of a patient's complete CT scan. We retrieved the 'PatientID' and 'Z-axis' coordinates from the metadata of each DICOM file and utilized this data to arrange the CT slices of each patient in a sequential manner. This portrayed a two-dimensional depiction of the CT scan of each patient, resulting in a total of 18,938 CT scans. 20% of the CT scans were randomly selected for the test dataset, leaving 15,154 CT scans for training and 3,784 CT scans for testing.

In the training dataset, the Epidural Hematoma class had the least amount of CT slices, specifically 1350 slices. To address this, we duplicated each epidural slice exactly once, resulting in 2,700 slices. We then randomly selected 2,700 slices from each hemorrhage category to form the 'Single' class, totaling 13,500 slices. To ensure a balanced training set, 13,500 CT slices were also randomly selected from both the Normal and Multiple classes. This ensured that all three training classes were evenly distributed with equal number of slices from each type of hemorrhage, without any bias. Finally, we applied preprocessing techniques, detailed in the next section, and converted the DICOM files to Portable Network Graphics (PNG) format for training and evaluation. A total of 40,500 CT slices were used for training and 128,320 slices were allocated for the test set.

III. METHOD

A. Preprocessing

Medical CT scans are typically stored in the DICOM format, containing both pixel data and metadata [23]. Our dataset consists of 16-bit images, where each pixel value corresponds to tissue density measured in Hounsfield Units (HU). Radiologists use specific HU intensity ranges to analyze different tissues and diseases, defined by a central value (WC) and range (WW), which are essential for diagnostic imaging [15]. Based on expert radiologist guidelines for detecting ICH, we use three HU windows: brain window (WC=40, WW=80), subdural window (WC=80, WW=200), and soft tissue window (WC=40, WW=380) [27]. These CT slices are merged into a single three-channel image, converting the grayscale slice into an RGB representation, where each color channel highlights different brain regions. As shown in Fig. 1, the red channel highlights brain tissue, the green channel highlights the

subdural tissues, and the blue channel highlights soft tissue [16]. In order to handle the extensive dataset, we resized the spatial dimension of the original 512×512 pixel CT slices to 224×224 pixels using linear interpolation, processing them in batches of 16.

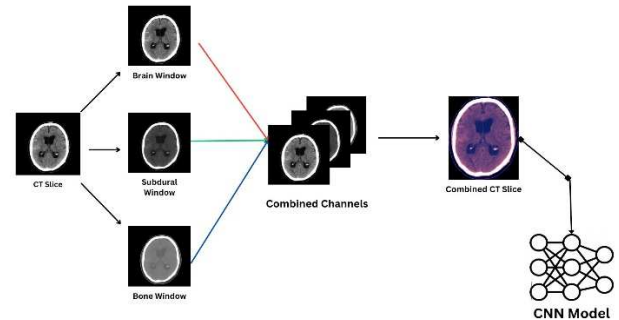


Fig. 1. Combination of window channels to form RGB CT slice

The CT slices underwent additional segmentation to isolate the skull region, removing unnecessary features and enhancing the model's training efficiency, as shown in Fig. 2. Initially, each slice was analyzed to identify the contour with the largest area, representing the skull. This was done by converting the slice to grayscale, applying a binary threshold, and using the external contour retrieval method [26] to find the contours. The contour area and the total slice area were calculated, and slices with a ratio below 15% were excluded. This filtering process eliminated slices that were significantly deformed or contained only the metal cover and other extraneous elements. After detecting the contours, a rectangular boundary was created around the skull area, and the rest of the slice was cropped out. This resulted in CT slices that only displayed the skull region with minimal black areas and unnecessary details, as illustrated in Fig. 3.

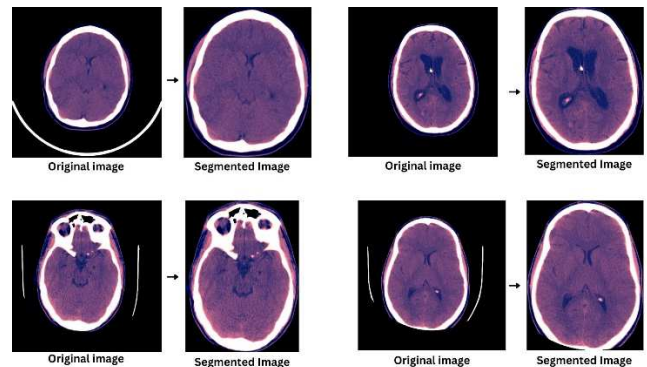


Fig. 2. Segmentation of skull region

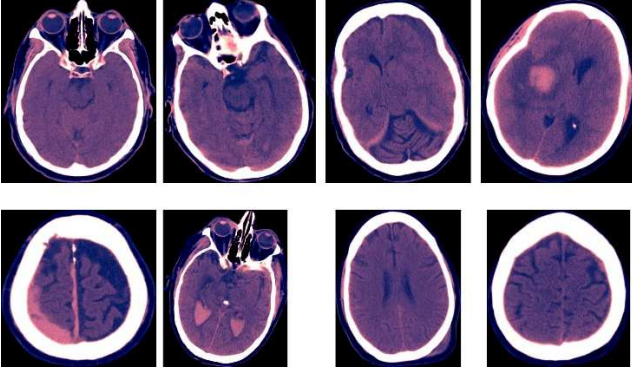


Fig. 3. Sample of CT slices used for training

B. Model

Each model utilized a pre-trained Convolutional Neural Network (CNN) backbone, with a global average pooling layer, along with two dense layers of 512 and 256 neurons, followed by batch normalization and ReLU activation layers [18]. To prevent bias, all models were constructed with the same architecture and trained using consistent initial and fine-tuning hyperparameters [21].

During initial training, each model underwent 50 epochs of training with a learning rate of 0.001, using the categorical cross-entropy loss function [26] and the Adam optimizer [19]. Two callbacks were implemented: ModelCheckpoint, to save the best-performing model based on validation loss, and ReduceLROnPlateau [20], to reduce the learning rate by 10% if validation accuracy plateaued for 5 consecutive epochs.

The model with the lowest validation loss was further trained for 80 more epochs at a reduced learning rate of 0.000001. The same loss function, optimizer, and evaluation criteria were maintained.

The top-performing models from both phases were then evaluated on the test dataset using metrics such as accuracy, precision, recall, and F1-score, which are commonly used in classification problems [22]. Additionally, a Receiver Operating Characteristic (ROC) curve and Grad-Cam heatmaps were also generated for each model based on the test set [17].

C. Evaluation

The categorical cross-entropy loss function, L , is described by (1). In this equation, N represents the number of samples, C represents the number of classes, and y_{ij} is a binary indicator that takes the value of 0 or 1, indicating whether class label j is the proper classification for sample i . The variable p_{ij} represents the predicted probability that the (i) th sample belongs to the (j) th class, whereas the term ‘log’ refers to the natural logarithm.

$$L = - \sum_{i=1}^N \sum_{j=1}^C y_{ij} \log(p_{ij}) \quad (1)$$

To calculate accuracy (2), precision (3), recall (4), and F1-score (5), we first determine the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). TP and TN represent correctly predicted images, while FP and FN represent incorrectly predicted ones. Precision measures the accuracy of positive predictions, while recall indicates the rate of correctly predicted positive instances among all actual positives. The F1-score is the harmonic mean of precision and recall.

The AUROC value (6) ranges from 0 to 1, where TPR_i and TPR_{i+1} are the true positive rates at consecutive points, while FPR_i and FPR_{i+1} are the false positive rates at consecutive points. These metrics are calculated using the following equations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (5)$$

$$AUROC = \sum_{i=1}^{n-1} \left(\frac{TPR_{i+1} + TPR_i}{2} \right) \times (FPR_{i+1} - FPR_i) \quad (6)$$

Through the utilization of this framework, we successfully assessed the performance of each model and derived significant insights on their respective advantages and disadvantages.

IV. RESULTS AND DISCUSSION

As shown in Table I and II, EfficientNetB4 outperformed InceptionV3 and DenseNet201 with higher accuracy, precision, recall, and F1-score in both the initial and fine-tuning phases. This can be attributed to EfficientNetB4’s advanced architectural features, including compound scaling, mobile inverted bottleneck (MBConv) layers, and squeeze-and-excitation (SE) layers [24].

TABLE I. INITIAL TRAINING METRICS

Model Names	Initial Training			
	Accuracy	Precision	Recall	F1-Score
InceptionV3	0.72	0.81	0.63	0.71
EfficientNetB4	0.78	0.83	0.72	0.77
DenseNet201	0.64	0.73	0.54	0.62

TABLE II. FINE-TUNED METRICS

Model Names	Fine Tuning Stage			
	Accuracy	Precision	Recall	F1-Score
InceptionV3	0.84	0.86	0.81	0.84
EfficientNetB4	0.85	0.87	0.84	0.86
DenseNet201	0.84	0.86	0.83	0.84

Compound Scaling is a key advantage of EfficientNetB4, which scales the network's depth, width and resolution in a balanced manner. This allows the model to capture complex patterns more effectively by combining deeper layers for hierarchical feature extraction, wider layers for capturing diverse features, and higher input resolutions for detecting fine details. This balanced approach enables the model to learn complex patterns in the data more efficiently without additional computational costs.

MBConv layers in EfficientNetB4 reduce the number of operations required while maintaining high performance. Unlike traditional bottleneck layers that compress and then expand feature channels, MBConv layers first expand the channels, apply depth-wise convolutions, and then reduce the channels back down. This allows EfficientNetB4 to process information in a more compact manner, further reducing computational costs while maintaining important features.

SE Layers enhance the model's focus on crucial information by generating weights that highlight the importance of each channel, improving its ability to detect subtle patterns and reduce overfitting.

In contrast, InceptionV3 and DenseNet201 lack these advanced features. InceptionV3 relies on inception modules and factorized convolutions to capture features at different scales, but it lacks the dynamic scaling and recalibration capabilities of EfficientNetB4. On the other hand, DenseNet201 uses dense connectivity, where each layer receives input from all previous layers, promoting feature reuse and gradient flow. However, it lacks the balanced scaling and efficient layer designs of EfficientNetB4.

This performance difference is illustrated in Fig. 4, Fig. 5, and Fig. 6. EfficientNetB4 shows a relatively small difference of 4% between training and validation accuracy, indicating good generalization within an acceptable range. In contrast, InceptionV3 has a 6% difference, and DenseNet201 has a 10% difference, showing clear signs of overfitting.

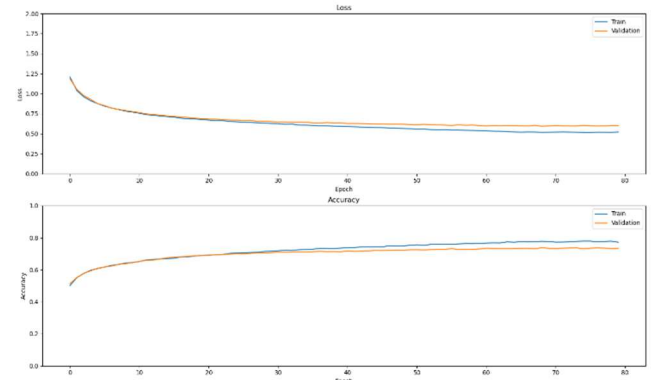


Fig. 4. Plot history of EfficientNetB4

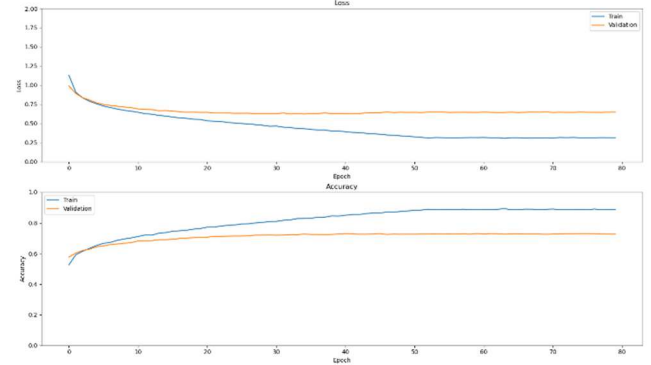


Fig. 5. Plot history of DenseNet201

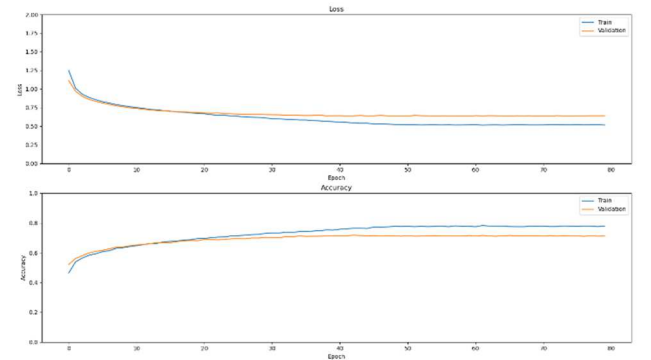


Fig. 6. Plot history of InceptionV3

These results are consistent with previous studies that highlight EfficientNetB4's strong performance in image classification, particularly in tasks that involve detecting subtle or complex patterns [25]. EfficientNetB4's slightly higher performance in accuracy, precision, and F1-score may seem small at first, but it can have a significant impact in practice. Even minor improvements can lead to the identification of more positive cases with greater accuracy, leading to earlier diagnoses and timely treatments. These advancements have the potential to save many lives in real-world medical settings. Additionally, The ROC curves in Fig. 7 and Grad-CAM heatmaps in Fig. 8 further confirm EfficientNetB4's ability to accurately identify potential hemorrhages, even when visual signs are not immediately obvious. The Grad-CAM heatmaps ensure the model focuses on the relevant areas for ICH detection, further proving its reliability.

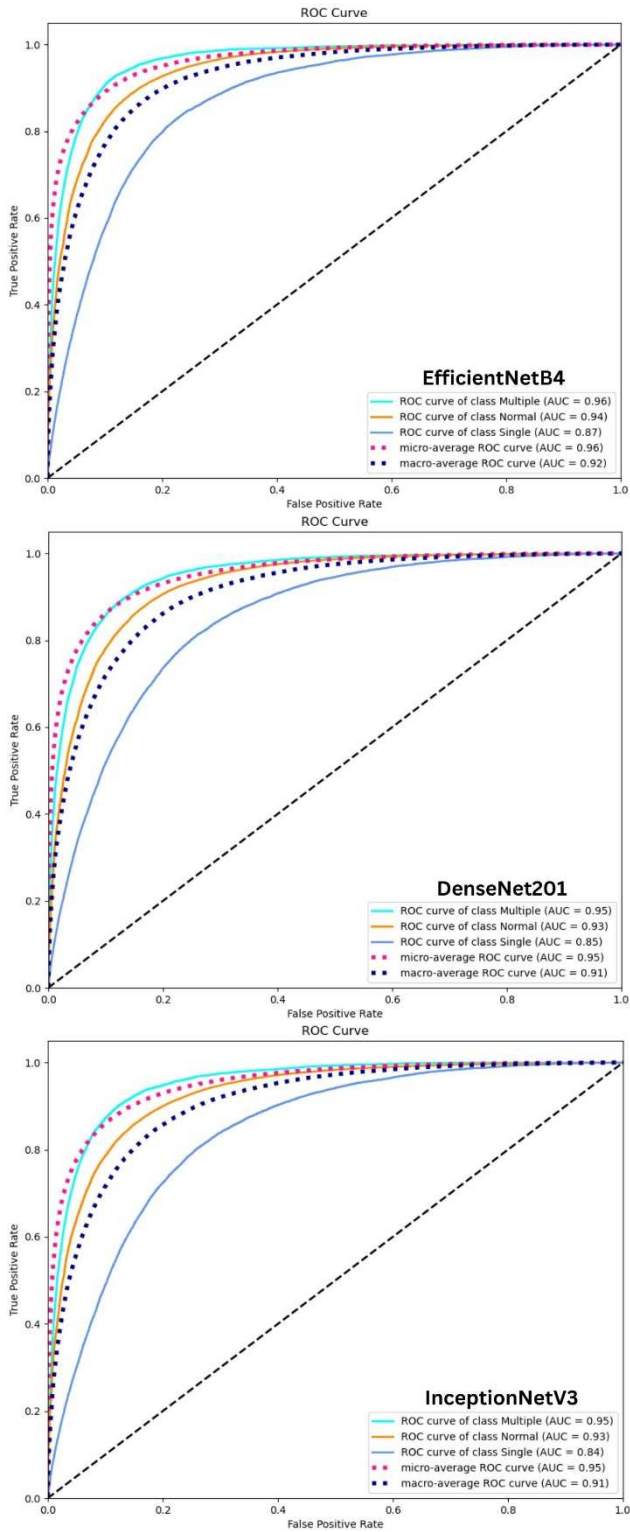


Fig. 7. ROC curve of all models

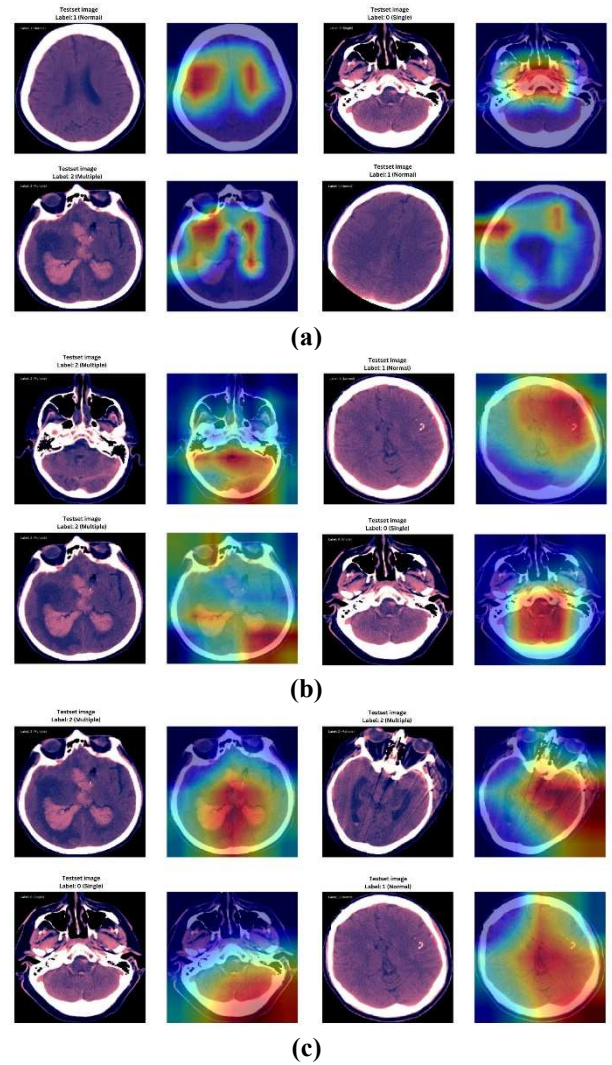


Fig. 8. Grad Cam heatmaps on test set, (a) EfficientNetB4, (b) DenseNet201, (c) InceptionV3

In summary, EfficientNetB4 stands out as the best model for complex classification tasks because of its efficient design and ability to scale well. While InceptionV3 and DenseNet201 are strong models as well, they do not match the efficiency and feature improvements that make EfficientNetB4 so effective. As a result, EfficientNetB4 performs better in all key areas, making it a great choice for situations where both high performance and real-time use are important. In clinical settings, this can lead to better patient outcomes and greater trust in AI tools for everyday diagnostics.

V. LIMITATIONS AND FUTURE WORK

The experiments were conducted on a desktop computer equipped with an Intel® Core™ i5-7500 Central Processing Unit (CPU) operating at a clock speed of 3.40 GHz. The system was running a 64-bit operating system and featured an x64-based processor. It was also equipped with an NVIDIA GeForce RTX 3060 Ti graphics card, which includes 4,864 Compute Unified Device Architecture (CUDA) cores, 152 Tensor cores, and 38 Ray Tracing (RT) cores. Additionally, the computer had 16 gigabytes (GB) of Random Access

Memory (RAM). All coding and development tasks were performed using Microsoft Visual Studio.

The main constraints in our study were limited computational resources and training time, which restricted us to using only 40,500 out of the total 752,602 CT slices for training and validation, which is only 5% of the total available dataset, while 128,320 slices were reserved for testing. These limitations may have affected the models' overall performance. However, our study still provides valuable insights by integrating advanced preprocessing techniques along with detailed comparative analysis of the three deep learning models, establishing a strong foundation for future research aimed at improving ICH detection.

In future work, we plan to increase the training dataset size and expand the number of classes to identify all hemorrhage categories, enhancing pattern recognition for more accurate diagnoses and better patient outcomes.

REFERENCES

- [1] M. Diringer, "Intracranial Hemorrhage," in *Bradley's Neurology in Clinical Practice*, 7th ed., A. Aminoff, J. Daroff, G. Gronseth, and M. J. S., Eds. Philadelphia, PA, USA: Elsevier, 2020, pp. 1515-1522.
- [2] J. A. Caceres and J. N. Goldstein, "Intracranial hemorrhage," *Emergency Medicine Clinics of North America*, vol. 30, no. 3, pp. 771-794, Aug. 2012.
- [3] Hemphill, J. C., 3rd, Greenberg, S. M., Anderson, C. S., Becker, K., Bendok, B. R., Cushman, M., Fung, G. L., Goldstein, J. N., Macdonald, R. L., Mitchell, P. H., Scott, P. A., Selim, M. H., Woo, D., American Heart Association Stroke Council, Council on Cardiovascular and Stroke Nursing, & Council on Clinical Cardiology (2015). Guidelines for the Management of Spontaneous Intracerebral Hemorrhage: A Guideline for Healthcare Professionals From the American Heart Association/American Stroke Association. *Stroke*, 46(7), 2032-2060
- [4] D. Guo, F. Wang, and S. Zhang, "Intracerebral hemorrhage: From clinical settings to animal models," *Int. J. Mol. Sci.*, vol. 19, no. 8, pp. 22-53, 2018.
- [5] S. J. An, T. J. Kim, and B.-W. Yoon, "Epidemiology, risk factors, and clinical features of intracerebral hemorrhage: An update," *J. Stroke*, vol. 19, no. 1, pp. 3-10, Jan. 2017
- [6] R. D. Rajashekar and J. W. Liang, "Intracerebral Hemorrhage," StatPearls [Internet], Treasure Island, FL: StatPearls Publishing, Feb. 6, 2023. [Online]. Available: <https://www.ncbi.nlm.nih.gov/books/NBK553103>
- [7] D. Woo et al., "Risk Factors Associated With Mortality and Neurologic Disability After Intracerebral Hemorrhage in a Racially and Ethnically Diverse Cohort," *JAMA Netw Open*, vol. 5, no. 3, p. e221103, 2022
- [8] J. S. Balami and A. M. Buchan, "Complications of intracerebral haemorrhage," *The Lancet Neurology*, vol. 11, no. 1, pp. 101-118, 2012
- [9] Grewal, M.; Srivastava, M.M.; Kumar, P.; Varadarajan, S. RADnet: Radiologist level accuracy using deep learning for hemorrhage detection in CT scans. In *Proceedings of the 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018)*, Washington, DC, USA, 4-7 April 2018; pp. 281-284
- [10] H. Lee, S. Yune, M. Mansouri, M. Kim, S. H. Tajmir, C. E. Guerrier, S. A. Ebert, S. R. Pomerantz, J. M. Romero, S. Kamalian, R. G. Gonzalez, M. H. Lev, and S. Do, "An explainable deep-learning algorithm for the detection of acute intracranial haemorrhage from small datasets," *Nature Biomedical Engineering*, vol. 3, no. 3, pp. 173-182, 2019
- [11] S. Chilamkurthy, R. Ghosh, S. Tanamala, M. Biviji, N. G. Campeau, V. K. Venugopal, V. Mahajan, P. Rao, and P. Warier, "Development and Validation of Deep Learning Algorithms for Detection of Critical Findings in Head CT Scans," *ArXiv*, 2018
- [12] H. Ye, F. Gao, Y. Yin, D. Guo, P. Zhao, Y. Lu, X. Wang, J. Bai, K. Cao, Q. Song, H. Zhang, W. Chen, X. Guo, and J. Xia, "Precise diagnosis of intracranial hemorrhage and subtypes using a three-dimensional joint convolutional and recurrent neural network," *Eur. Radiol.*, vol. 29, no. 11, pp. 6191-6201, 2019
- [13] Y. H. Li, L. Zhang, Q. M. Hu, H. W. Li, F. C. Jia, and J. H. Wu, "Automatic subarachnoid space segmentation and hemorrhage detection in clinical head CT scans," *Int. J. Comput. Assist. Radiol. Surg.*, vol. 7, no. 4, pp. 507-516, 2012
- [14] A. Stein, M.D., C. Wu, C. Carr, G. Shih, J. Kalpathy-Cramer, J. Elliott, L. Prevedello, M. Kohli, M.D., M. Lungren, P. Culliton, R. Ball, and S. Halabi, M.D., "RSNA Intracranial Hemorrhage Detection," *Kaggle*, 2019
- [15] Hosny A, Parmar C, Quackenbush J, Schwartz LH, Aerts HJWL. Artificial intelligence in radiology. *Nat Rev Cancer*. 2018 Aug;18, pp. 500-510
- [16] A. Saha and R. G. Abramson, "Role of Imaging in Intracranial Hemorrhage," *Neuroimaging Clinics of North America*, vol. 29, no. 2, pp. 147-157, 2019
- [17] M. Provost, F., & Fawcett, T. (2001). Robust classification for imprecise environments. *Machine Learning*, 42(3), 203-231
- [18] A. Karpathy, "Convolutional Neural Networks for Visual Recognition," CS231n: Convolutional Neural Networks for Visual Recognition. [Online]. Available: <https://cs231n.github.io/convolutional-networks/>
- [19] Kingma, D. P., & Ba, J. (2014, December 22). Adam: A method for stochastic optimization. *arXiv.org*. <https://arxiv.org/abs/1412.6980>
- [20] L. N. Smith, "Cyclical Learning Rates for Training Neural Networks," *2017 IEEE Winter Conference on Applications of Computer Vision (WACV)*, Santa Rosa, CA, USA, 2017, pp. 464-472
- [21] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proceeding IEEE Conf. Computer Vision and Pattern Recognition*, 2017, pp. 2261-2269
- [22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceeding IEEE Conf. Computer Vision and Pattern Recognition*, 2016, pp. 770-778
- [23] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proceeding IEEE Conf. Computer Vision and Pattern Recognition*, 2019, pp. 4510-4519
- [24] Phaphuangwittayakul A, Guo Y, Ying F, Dawod AY, Angkurawaranon S, Angkurawaranon C. An optimal deep learning framework for multi-type hemorrhagic lesions detection and quantification in head CT images for traumatic brain injury. *Appl Intell (Dordr)*. 2022;52(7):7320-7338. Epub 2021 Sep 25. PMID: 34764620; PMCID: PMC847537
- [25] Yang Y, Zhang L, Du M, Bo J, Liu H, Ren L, Li X, Deen MJ. A comparative analysis of eleven neural networks architectures for small datasets of lung images of COVID-19 patients toward improved clinical decisions. *Comput Biol Med*. 2021 Dec;139:104887. Epub 2021 Sep 24. PMID: 34688974; PMCID: PMC8461289
- [26] Tan, M., & Le, Q., V. (2019, May 28). EfficientNet: Rethinking model scaling for convolutional neural networks. *arXiv.org*. <https://arxiv.org/abs/1905.11946>
- [27] S. Suzuki and K. Be, "Topological structural analysis of digitized binary images by border following," *Computer Vision Graphics and Image Processing*, vol. 30, no. 1, pp. 32-46, Apr. 1985